

The examination is being carried out on the **following application documents**

Description, Pages

1-305 as published

Claims, Numbers

1-15 filed in electronic form on 07-10-2022

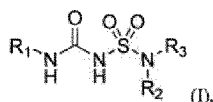
1 Amendments (Article 123(2) EPC)

The amendments filed with the letter of 07.10.2022 are considered to fulfill the requirements of Article 123(2) EPC.

2 Inventive step (Article 56 EPC)

The objections raised under Article 56 EPC with the previous communication are maintained.

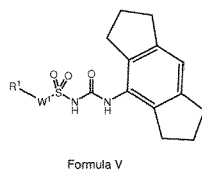
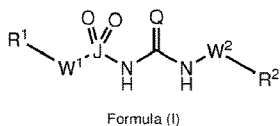
2.1 The present claimed compounds of formula (I) are useful to treat diseases connected to the inhibition of the NLRP3 activation.



wherein R1 is a cycloalkyl or aryl, R2 is -(CX1X2)_n-R2s with n=0-2 and R2s=heterocycloalkyl and R3 is a heterocycloalkyl or heterocycloaryl.

2.2 Closest prior art

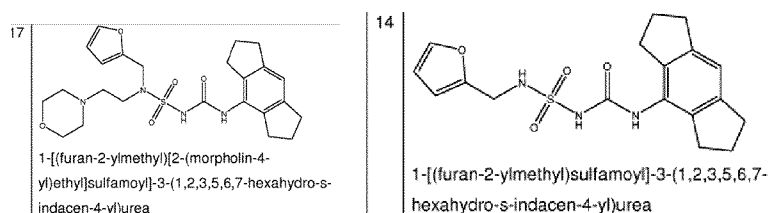
D1, which is regarded as being the closest prior art, discloses compounds of formula (I) (claim 1) or (V)(claim 7) with the same uses as the present ones.



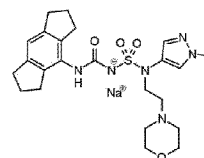
The subject-matter of claim 1 can be considered as a selection of a formula (I) from the compounds of formula I or V disclosed in this document (claims 1 and 7) for the same use. The main selection which appears to be the core of the

current claimed subject-matter is represented by the definitions of the present R2 and R3 which contain both a heterocycle moiety, wherein one of them (R3) directly linked on the N atom.

As regards the specific examples disclosed in D1, compounds 14, 17, 19, 35, 39 and the ones disclosed on page 30 appear to be the closest structural for the present ones.



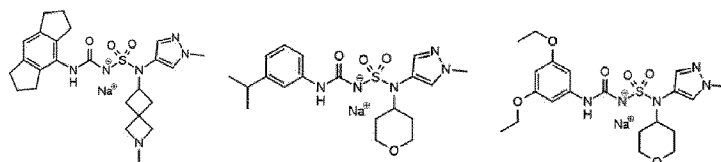
With the letter of 07.10.2022, the Applicant filed comparative data between



compound 17 of D1 and present example 46:

The Applicant argued that example 17 of D1 has in the PBMC assay higher than 10µm while the present example 46 has an IC50 lower than 1µm. This argument might be relevant for assessing inventive step if would have been credible that all claimed compounds posses the claimed improvement.

Considering the specific data provided on pages 301 to 116 of the current specification, it can be seen that compounds 58, 71 and 90 have also an IC50 in the range of 10 to 50µm:

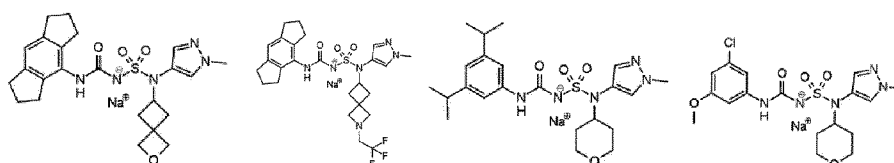


comp 58

comp 71

comp 90

while examples 66, 73 and 81 are disclosed to have an IC50 between 1 to 3µm, example 69 between 3 and 10 µm, and example 93 less than 1 µm:



In view of the very small structural differences between the compound 58, 71 and 90, which have an IC₅₀ higher than 10µm, and the closest structural compounds 66, 73, 81, and 69 which have an IC₅₀ lower than 10µm it cannot be concluded which is the differentiating feature responsible for the improved activity.

The same variation in activity due to small structural differences could also be seen from the data provided for the D1 compounds on pages 128-129.

As regard the argument provided by the Applicant with the letter of 07.10.2022 that none of the specific examples disclosed in D1 contains a directly attached heterocycle on the nitrogen atom, the Examining Division would like to underline that there are several examples in D1 containing a directly attached heterocycle (without having a second substituent on the N atom) and having a very good activity.

Thus, it is considered that since the present general formula claimed by claim 1 is a selection over the structure (I) of D1 and D1 discloses examples having a heterocycle directly linked on the nitrogen corresponding to the present NR₂R₃ as well as compounds wherein the nitrogen corresponding to the present NR₂R₃ is bisubstituted, the skilled person would expect the same level of activity for the present claimed compounds. Combining the bisubstitution on the nitrogen corresponding to the present NR₂R₃ (as disclosed for some examples of D1) with the presence of a directly linked heterocycle (as disclosed for other examples in D1, even if they contain a H as a second substituent on the above quoted N), it cannot be considered as inventive per se.

The data used by Applicant as comparison in the letter of 07.10.2022 cannot, in view of the experimental data provided in the present application and in D1, support the whole claimed scope.

The Applicant is invited to file comparative data which would support the whole claimed scope. In the case that it is not credible that all claimed compounds would possess the claimed improvement and since for a novel selection an inventive step could be acknowledged only on account of an unexpected effect over the compounds disclosed in D1, summons to oral proceedings might be issued (see also the objections raised in the previous communications).